

Gamblers or Delegates: Identifying Hidden Participant Roles in Crypto Casinos

Jiaxin Wang¹, Qian'ang Mao¹, Hongliang Sun², Jiaqi Yan^{1*}

1. School of Information Management, Nanjing University, China

2. University of Nottingham Ningbo China

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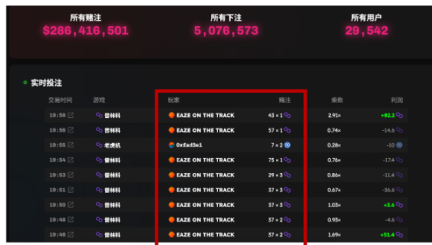
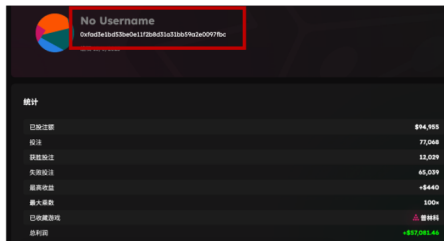
Background: What is Crypto Gambling?

- Crypto casinos provide gambling games (e.g., poker, blackjack, roulette, slot machines) on multi-chain blockchain networks (e.g., Ethereum, EOS, TRON, Arbitrum).
- Crypto casinos utilize cryptocurrencies or issue their own tokens, which can be used for placing bets, receiving payouts, and participating in platform governance.
- In some crypto casinos, gambling funds or tokens strictly follow predefined rules set by smart contracts to ensure perceived fairness and transparency of game results, addressing issues like fictitious prize pools, opaque processes, high costs, and unpaid winnings.



Background: What is Crypto Gambling?

- **Account-based:** Users create an account in DApps, deposit cryptocurrencies, play games, and withdraw winnings.
- **Wallet-based:** Players connect their cryptocurrency wallets directly to the platform. Transactions are made directly from the wallet.
- **Hybrid model:** Hybrid models combine both approaches, offering flexibility.



Reality: Involved in Grey and Black Market Industries

- However, the very **anonymity** afforded by blockchain technology renders crypto casinos **less than absolutely fair and transparent**, exposing them to **security and compliance** risks. [McKenzie and Silverman(2023)].
- According to reports from the United Nations, cryptocurrencies and casinos have emerged as major facilitators of **money laundering, underground banking, and cyber fraud** in East and Southeast Asia, often occupying **the gray or black crypto market sectors**.



Inspirations: *Delegatee* Roles in Traditional Casinos

Traditional Casinos

- A widely acknowledged issue is that a small number of behind-the-scenes **insiders or shills control and operate** the entire casino in an opaque manner [Trowbridge(1995)].
- These roles, called **Delegatees**, are crucial in undertaking diverse tasks such as prize management and project promotion for casinos [Wood and Griffiths(2015)].



Figure: Delegatee Roles in Traditional Casinos

Motivation and Purpose

- **Motivation:** A similar but more challenging situation occurs in crypto casinos. These delegates exist hidden among gamblers and even dispersed across large-scale anonymous networks, making them more difficult to identify.
- **Negative Impacts:** Misleading players, manipulating the market, committing fraud.
- **Purpose:** To pinpoint **hidden delegates** across large-scale and anonymous transaction networks, offering authorities insights for **better regulation** of crypto casinos.

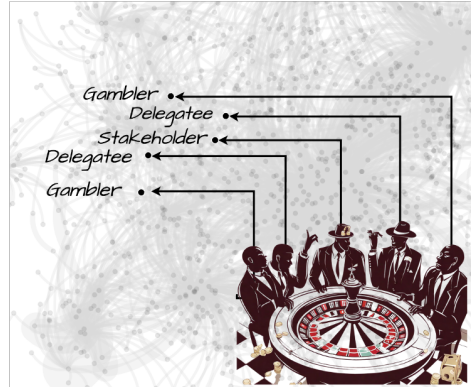


Figure: Who are *Gamblers* or *Delegates*?

Related Work: Crypto Gambling

- Mainly focusing on **psychological and socioeconomic** issues.
 - Betting data on Ethereum is analyzed from a decentralized application through probability-theoretic models [Meng and Fu(2020)].
 - [Scholten(2022)] analyze actual transaction data gathered from several decentralized gambling applications, providing a first look at engagement and behavioral groups.
- **Computer science research** has recently emerged.
 - [Wang et al.(2023)] proposed a novel detection approach based on GNNs to identify crypto gambling addresses and characterize their interaction patterns.
 - [Huang et al.(2022)] automatically identified contracts and crypto addresses involved in gambling by smart contract codes and address transaction records.
- **Research Gap 1: There remains a scarcity of research that delves into the operation of crypto casinos.**

Related Work: Crypto-crime Regulation and Forensics

- **Descriptive statistics** [Chen et al.(2020)]
- **Account classification** [Liu et al.(2022), Zhou et al.(2022)]
- **Anomaly detection** [Chen et al.(2018), Zhou et al.(2024), Wu et al.(2021)]
- **Transaction tracing** [Wu et al.(2023), Lin et al.(2022)]
- **Research Gap 2:**
 - These techniques mainly focus on **address-level identification** in extremely limited labeled scenarios. In reality, most anonymous addresses lack explicit labels, leading to an extreme imbalance with unlabeled addresses.
 - Existing address labels typically indicate **the types of transaction scenarios** entities are involved in, such as exchanges, gambling, scams, DeFi, and mining, without delving further into **the specific roles** these entities play in their respective scenarios.

Related Work: Role Identification

- Role identification is determined based on an individual's position, behavioral patterns, and relationships with other nodes to ascertain their function or status within the system.
- Intuitively, if two nodes have **similar structures**, they belong to the same role.
- Unlike community detection tasks, role identification focuses on the structure, function, and status of nodes within the entire network, rather than just locally dense clusters [Henderson et al.(2012)].

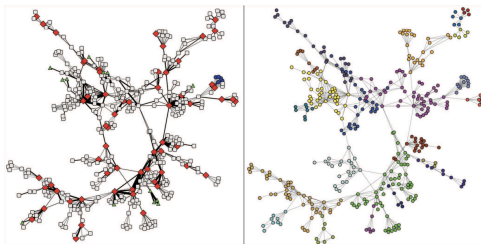
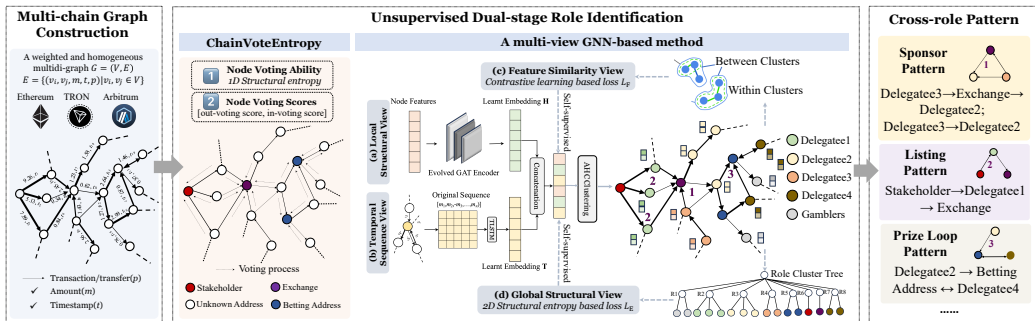


Figure: Role Recognition VS Community Detection

Related Work: Role Recognition

- Role identification can be divided into two types: **explicit role identification** and **implicit role identification**
 - **Explicit role identification** involves predefined specific roles based on existing theories or experience, and typically uses **supervised or semi-supervised learning** methods for classification.
 - **Implicit role identification** does not require prior definitions; traditionally, it has relied on statistical analysis and low-rank approximation, but recently, **unsupervised learning methods** have been commonly used.
- **Research Gap 3**
 - They are easily affected by the inherent randomness and noise of the graph, leading to a lack of robustness
 - Most focus only on local network structures while ignoring global information
 - Domain-specific knowledge has not been fully utilized, resulting in a lack of interpretability.

Methodology: Overview

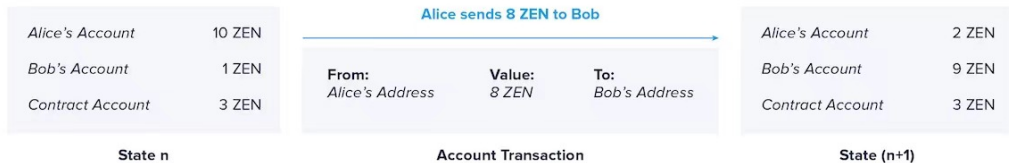


- **Multi-chain Graph Construction:** To construct a temporal, weighted, and directed graph model.
- **Dual-stage Role Identification:** To design a dual-stage role identification method, including a node voting-based indicator for **key roles** and a GNN-based framework for **hidden delegates**.
- **Cross-role Pattern Analysis:** To carry out a cross-role network motif extraction method to understand collusion or covert cooperation among roles.

Module 1: Multi-chain Graph Construction

Current crypto gambling activities mainly rely on public blockchain systems based on the **account model** such as Ethereum, TRON, and Arbitrum [Brünjes and Gabbay(2020)].

- Unlike the UTXO model of Bitcoin, an address in the account model can be categorized into two categories: **Externally Owned Account (EOA)** and **Contract Account (CA)**.
- An EOA refers to an account or address controlled by an external entity, typically an individual or organization. CAs are controlled by smart contracts and are created by EOAs or other CAs.
- Interactions between accounts involve the transfer of crypto assets (e.g., ETH, TRX, tokens). **Each transaction has a single input and a single output address.**



Module 1: Multi-chain Graph Construction

Definition 1. (Original Graph Model, og-IGM):

A weighted and heterogeneous multidigraph $G = (V_{EOA}, V_{CA}, E_t, E_c)$

- V_{EOA} and V_{CA} represent EOAs and CAs.
- $E_t = \{(v_i, v_j, a, t) \mid v_i, v_j \in V_{EOA}\}$ represents the directed edge set constructed from transaction information.
- $E_c = \{(v_i, v_j, f, t) \mid v_i, v_j \in V_{EOA} \cup V_{CA}\}$ represents the directed edge set constructed from token transfer.
- Edge attributes a , t and f represent transaction amounts, timestamps, and token values, respectively.

Definition 2. (Lightweight Graph Model, lw-IGM):

A weighted and homogeneous multidigraph $G = (V, E)$

- V represents the set of all addresses
- Each node $v \in V$ has an extra attribute that indicates whether it is an EOA or CA.
- $E = \{(v_i, v_j, m, t, p) \mid v_i, v_j \in V\}$ represents interactions between addresses.
- Edge attributes m , t , and p represent the interaction amount, timestamp, and whether the interaction is a transaction or token transfer, respectively.

Module 2: Why was it designed in **Dual-stage**?

Why is Dual-stage (Key Role Identification + Hidden Role Identification) ?

- In real-world networks, some nodes exhibit atypical features due to **noise or anomalies**.
- Hidden roles are often not immediately apparent, and directly analyzing all nodes increases the complexity of the task.
- Key nodes, which have greater influence and dissemination capabilities within the network, can be identified first to reduce redundant information.
- This provides more precise guidance and reference for subsequent hidden role identification, improving the overall efficiency and accuracy of the process.

Module 2: Key Role Identification-*ChainVoteEntropy*

- **Why is ChainVoteEntropy?** — Traditional centrality measures fail to fully capture the dynamic interaction pattern and structure complexity of blockchain networks.
- **Why is Vote-style?** — The transaction behavior observed in token issuance and transfer can be seen as the inverse of node voting in networks.
- **Why is Entropy?** — See the next two pages
- Combining these two aspects, we propose a novel indicator named **ChainVoteEntropy**, designed for identifying key roles in directed and weighted blockchain transaction networks.

Module 2: Key Role Identification-*ChainVoteEntropy*

(i) Voting scores.

- Different roles vary significantly in terms of directionality.
 - For instance, *Exchanges* and *Service* often handle large volumes of both incoming and outgoing transactions.
 - *Stakeholders* are typically involved in outgoing transactions.
- Each node v obtains its in-voting score $\tilde{N}_v^{\text{in}}$ and out-voting score $\tilde{N}_v^{\text{out}}$ separately from its input and output neighbors.
- Voting scores of v are influenced by the number of neighbors and the weighted sum of voting ability scores from neighbors of v , are defined as:

$$\tilde{N}_v^{\text{in}}(t) = \sqrt{\left(\sum_{(u,v)} b_u^{\text{out}}(t) \cdot m_{(u,v)}\right) \left|\tilde{E}_v^{\text{in}}\right|} \quad (1)$$

$$\tilde{N}_v^{\text{out}}(t) = \sqrt{\left(\sum_{(v,u')} b_{u'}^{\text{in}}(t) \cdot m_{(v,u')}\right) \left|\tilde{E}_v^{\text{out}}\right|} \quad (2)$$

Module 2: Key Role Identification-*ChainVoteEntropy*

(ii) Voting ability.

- The voting ability of a node can be regarded as its capacity to disseminate information or influence other nodes.
- **1D Structural Entropy** provides an effective metric for measuring voting ability.
- Nodes with higher 1D SE have more connections across a larger and more diverse neighbors, resulting in more information dissemination paths and greater voting ability.
- Therefore, let each node $(\tilde{N}_v^{\text{in}}(t), \tilde{N}_v^{\text{out}}(t), b_v^{\text{in}}(t), b_v^{\text{out}}(t))$ is initialized as $(0, 0, H_v^{\text{in}}, H_v^{\text{out}})$, where H_v^{in} and H_v^{out} represent the 1D in- and out- structural entropies of v respectively, as defined below.

$$b_v^{\text{in}}(0) = -\frac{wd_v^{\text{in}}}{V_G} \log_2 \frac{wd_v^{\text{in}}}{V_G}, b_v^{\text{out}}(0) = -\frac{wd_v^{\text{out}}}{V_G} \log_2 \frac{wd_v^{\text{out}}}{V_G} \quad (3)$$

- where wd_v^{in} and wd_v^{out} represent the weighted (amounts) in-degree and out-degree of v respectively. V_G represents the sum of weights of all edges in G .

Module 2: Key Role Identification-*ChainVoteEntropy*

(iii) Voting process.

- In each iteration, selected key addresses that score higher between the highest $\tilde{N}_v^{\text{in}}$ and $\tilde{N}_v^{\text{out}}$ do not participate in subsequent elections.
- For in-neighbors of the selected node, their out-voting ability decreases by $\frac{1}{\langle k \rangle}$, where $\langle k \rangle$ is the average degree of G .
- For out-neighbors of the selected node, their in-voting ability decreases by $\frac{1}{\langle k \rangle}$.
- Other voting ability values remain unchanged. This process repeats until p key nodes are selected.

(iv) Time Complexity.

- Blockchain transaction networks are typically **sparse graphs** ($E \sim O(N)$) [Bai et al.(2021)]
- *ChainVoteEntropy* primarily involves 1D structural entropy calculation and the voting process, with a **linear time complexity** $O((N + E) + p \cdot (N + E)) \sim O(N)$.

Module 2: Hidden Role Identification-Feature Extraction

- Once key roles are determined, the node sets for all paths between role pairs can be obtained.
- We extracted **manual statistical features for network structural and temporal evolution embeddings** to generate a comprehensive representation for each address.
- **Manual statistical features** are handcrafted based on domain knowledge and historical data through statistical methods, which help profile individual accounts and behaviors.
 - **Account state features** include account balance, bounce, etc.
 - **Transaction intensity features** encompass input/output transaction amounts, gas fees, etc.
 - **Transaction frequency features** relate to the number of input and output transactions within a specific period, etc.
 - Additionally, we derive **aggregated features** through operations such as summation, averaging, variance, Gini coefficient, etc.

Module 2: Hidden Role Identification-Statistical Features

Type	Symbol	Description
Account State	Balance	The amount of the cryptocurrency or token owned by the address
	Nounce	The total transaction amount of the address
Transaction Intensity	Total_amount	The total amount of all transactions of the address
	Total_in_amount	The total amount of input transactions of the address
	Total_out_amount	The total amount of output transactions of the address
	Avg_amount	The average amount of all transactions of the address
	Avg_in_amount	The average amount of input transactions of the address
	Avg_out_amount	The average amount of output transactions of the address
	Var_amount	The standard deviation amount of all transactions of the address
	Var_in_amount	The standard deviation amount of input transactions of the address
	Var_out_amount	The standard deviation amount of output transactions of the address
	Gini_amount	The Gini coefficient amount of all transactions of the address
	Gini_in_amount	The Gini coefficient amount of input transactions of the address
	Gini_out_amount	The Gini coefficient amount of output transactions of the address
Transaction Frequency	Gas	The transaction gas supplied by the sender of the address
	Gas_in	The gas supplied by the input transaction sender of the address
	Gas_out	The gas supplied by the output transaction sender of the address
	Num_all_tran	The number of all transactions of the address
	Num_in_tran	The number of input transactions of the address
	Num_out_tran	The number of output transactions of the address
	Fre_all_tran	The frequency of all transactions of the address
	Fre_in_tran	The frequency of input transactions of the address
	Fre_out_tran	The frequency of output transactions of the address
	R_in_out	The ratio of the number of input/output transactions of the address

Module 2: Hidden Role Identification-Structural/Temporal Embedding

- Two nodes with similar **structures** belong to the same role [Henderson et al.(2012)].
- We utilize an **Attributed Graph Attention Network** to combine edge attributes with the graph structure in the latent representation.
- Different transaction amounts with neighbors have different influences on target nodes.
 - High-value transactions may indicate stronger trust relationships or higher capital liquidity between nodes
 - Low-value transactions may signify more daily or trivial interactions.
- Transactions occurring at different times may represent different behavior patterns.
- **Temporal LSTM** model can be used to capture the evolution features. These transactions are sorted in ascending order by timestamp and then input into the model.

Module 2: Hidden Role Identification-Why needs SSL?

- The main challenge of the proposed method is **the absence of label guidance**, which means there is no feedback about whether the results are robust and well-optimized.
- To design a **self-supervised learning** based on **graph clustering** to **adaptively achieve representation learning with greater generalizability**.
 - To input the concatenated embedding into **Agglomerative Hierarchical Clustering (AHC)** [Bhattacharjee and Mitra(2021)].
 - Gambling transaction networks consist of time-ordered transaction event sequences, exhibiting evident hierarchical organizational structures between roles.
 - AHC algorithm, which adaptively determines the number of clusters, is suitable for datasets with hierarchical structure and uneven density distribution.

Module 2: Hidden Role Identification-SSL-Global Structure

- **Why?** Roles are collections of structurally similar entities, and currently, we only consider local neighborhood structures **without considering the influence of global structures**.
- To employ **2D structural entropy minimization** to enhance the robustness of role partitioning, while simultaneously capturing global structural information.
- Given a role partitioning result $R = \{R_1, R_2, \dots, R_L\}$ for G , we define:

$$\mathcal{L}_E = - \sum_{l=1}^L \frac{V_{R_l}}{V_G} \sum_{v \in R_l} \frac{wd_v^{in}}{V_{R_l}} \log_2 \frac{wd_v^{in}}{V_{R_l}} - \sum_{v \in V_{R_l}} \frac{g_l}{V_G} \log_2 \frac{V_{R_l}}{V_G} \quad (4)$$

- where L is the number of role clusters in R . wd_v^{in} is the weighted (amounts) in-degree of node v . V_{R_l} is the sum of the weighted in-degree of all nodes in R_l . g_l is the number of edges that cross cluster boundaries in R_l . V_G is the total weight of all edges in G .

Methodology: Hidden Role Identification-SSL-Feature Similarity

- **Why?** The absence of feedback about **whether clusters are well explained**.
- To design a self-supervised contrastive learning to achieve greater **interpretability**.
 - Intra-cluster loss, which minimizes the sum of squared Euclidean distances within the same cluster to bring similar samples closer.
 - Inter-cluster loss, which maximizes the sum of squared Euclidean distances between different clusters to push dissimilar samples farther apart.

$$\mathcal{L}_F = \frac{1}{|R_k|} \sum_{i,j \in R_k} \|\mathbf{z}_i - \mathbf{z}_j\|^2 - \frac{1}{|R_k||R_l|} \sum_{i \in R_k, j \in R_l} \|\mathbf{z}_i - \mathbf{z}_j\|^2 \quad (5)$$

- where R_k and R_l represent two different clusters, $|R_k|$ and $|R_l|$ represent their number of nodes respectively. $\mathbf{z}_i$ and $\mathbf{z}_j$ are the concatenated embedding of node i and j .
- Finally, we **jointly learn** two SSL losses as:

$$\mathcal{L}_{SSL} = \alpha \cdot \mathcal{L}_E + (1 - \alpha) \cdot \mathcal{L}_F \quad (6)$$

Methodology: Cross-role Pattern Analysis

- Interactions across various roles play a pivotal role in shaping the overall operation of organizations.
- Our objective is to delineate how these roles may engage in **collusion or covert cooperation**. (that is to identify **frequent interaction patterns**.)
- A cross-role network motif extraction algorithm:
 - To extract motifs that involve at least 2 distinct roles based on network motif mining.
 - Once cross-role motifs are extracted, their significance is determined based on their occurrence frequency and z-scores [Konagurthu and Lesk(2008)].

Algorithm 2 Cross-Role Network Motif Extraction

Input: Graph $G = (\tilde{V}, \tilde{E})$, Role labels $R = \{r_1, r_2, \dots, r_n\}$

Parameter: Node number of one motif $\tilde{S}$, Number of randomized networks N , Role number of one motif β , Threshold of z-score θ

Output: Cross-role network motifs $\tilde{M}$

```
1: Subgraphs  $\tilde{g} \leftarrow \emptyset$ 
2: for Each size  $s \in \tilde{S}$  do
3:    $\tilde{g}_s \leftarrow \text{FindSubgraphs}(G, s)$ 
4: Motifs  $\tilde{M} \leftarrow \emptyset$ 
5: for Subgraph  $g \in \tilde{g}$  do
6:   the set of roles:  $\tilde{R}_g \leftarrow \emptyset$ 
7:   for Node  $v \in g$  do
8:     if  $\text{role}(v) \notin \tilde{R}_g$  then
9:        $\tilde{R}_g \leftarrow \tilde{R}_g \cup \{\text{role}(v)\}$ 
10:  if  $|\tilde{R}_g| \geq \beta$  then
11:     $z \leftarrow \text{z-score}(g, N)$ 
12:    if  $z > \theta$  then
13:       $\tilde{M} \leftarrow \tilde{M} \cup \{g\}$ 
14: return Significant cross-role network motifs  $\tilde{M}$ 
```

Scenario Selection

This study focuses on three mainstream public blockchains: **Ethereum, TRON, and Arbitrum** to validate the effectiveness and generalizability of CCDM.

- Ethereum is currently the primary crypto platform.
- TRON, due to its low transaction fees, is the most popular blockchain for gambling activities to date.
- Emerging casinos primarily concentrate on Layer 2 solutions such as Arbitrum, which offers faster transaction speeds and lower fees.

Data Preparation

Experiments consist of **native transactions, token transfers, and address-label datasets**.

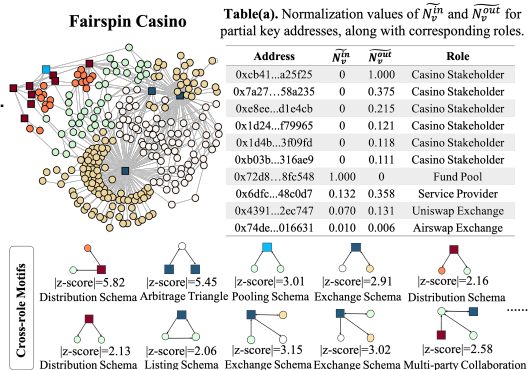
- To use third-party software *Erigon*, *java-tron*, and *Nitro* to synchronize the raw block data for Ethereum, TRON, and Arbitrum respectively.
- Then the open-source ETL tool *web3research* is used to parse the raw data.
- Labels of *crypto casinos* and *Exchanges* come from industry-trusted sources such as *etherscan*, *dappradar*, *tronscan*, and *arbiscan*.
- Other role labels are from manually verified labels by official law enforcement agencies in real-world criminal cases.

Baselines and Evaluation Metrics

- **Unsupervised graph representation learning:**
 - GNN-FiLM [Brockschmidt(2020)], SuperGAT [Kim and Oh(2022)], EGC [Tailor et al.(2022)], GATv2 [Brody et al.(2022)], DirGNN [Rossi et al.(2023)].
 - Next, the generated representation is input into K-means [Hartigan and Wong(1979)] and DBSCAN [Day and Edelsbrunner(1984)] for unsupervised clustering.
- **Role identification methods:** RoIX [Henderson et al.(2012)] is a scalable unsupervised learning approach for automatically extracting structural roles from general network data.
- **Web3 methods:** Evolved PageRank combined with Local Community Detection (EPLCD) [Wu et al.(2023)].
- **Purity, Accuracy(ACC), and Normalized Mutual Information(NMI)** are used.
 - Purity and NMI are typically used for evaluating clustering results.
 - These indicators depend on some known true label.
 - *Purity* measures the extent to which clusters contain a single class.
 - *ACC* assesses the match between predicted and true labels by the optimal one-to-one mapping.
 - *NMI* quantifies the shared information between true and predicted clusters.

Case Studies-Ethereum

- We select one of the most popular gambling tokens, *Fairspin Token*(0xc2A8...580883), as our Ethereum casino case.
- Both $\tilde{N}_v^{in}$ and $\tilde{N}_v^{out}$ values are very high, Identified as exchanges verified on *etherscan.io*.
- **Airdrop promoters:** Be involved in promoting and marketing gambling tokens.
- **Listing agents:** Serving as intermediaries between stakeholders and exchanges/services.
- **Arbitrageurs:** Market participants specializing in capitalizing on price differences of tokens across exchanges.
- **Real gamblers:** Actively participating in games with cryptocurrency, aiming for game-based earnings.



Case Studies-Ethereum

- Exchanges have the greatest influence within the network.
- Stakeholders primarily focus on token issuance, hence exhibiting a relatively higher proportion of outgoing connections.
- Arbitrageurs exhibit relatively high betweenness centrality, indicating their role as bridges within the entire network.
- Listing agents have a closeness centrality value of 0.30, which is the highest except for Exchanges, suggesting they occupy a more central position and are closer to other nodes.

Table(b). Network features of different roles, including *in-degree*, *out-degree*, *weighted in-degree*, *weighted out-degree*, *closeness centrality*, *betweenness centrality*, and *PageRank*

Type	Role/Average Metrics	d_{in}	d_{out}	wd_{in}	wd_{out}	<i>closeness</i>	<i>betweenness</i>	<i>PageRank</i>
Key Roles	 Casino Stakeholders	1.14	12.29	3.57E+08	7.14E+08	0.003	0.00036	0.0011
	 Exchanges or Services	42.50	52.63	8,654,029	5,167,423	0.220	0.09787	0.0450
Hidden	 Fund Pools	7.00	0.00	1.44E+09	0	0.204	0.00000	0.0057
	 Arbitrageurs	2.66	2.40	518,361	518,074	0.195	0.00629	0.0030
	 Listing Agents	1.89	1.66	612,082	607,262	0.0932	0.00147	0.0018
Delegates	 Airdrop Promoters	2.10	0.6	5.15E+07	1.82E+05	0.0052	0.00037	0.0012
Gamblers	 Regular Gamblers	1.89	1.16	491,229	421,968	0.1643	0.00133	0.0025

Case Studies-TRON

- Through Telegram chat groups, we obtained cross-border gambling promotional TRON-based websites **Fānhuā Hash**.
- "Fānhuā Hash"* operates by utilizing block hashes as the subject of speculation.
- The primary approach involves placing bets on TRX by transferring varying amounts to gambling addresses through crypto wallets.
- The platform supports the exchange of TRX and USDT.
- "Fānhuā Hash"* offers a range of **gambling rules, such as "Odd-Even Hash", "Lucky Hash", and "Bull Hash", each of which corresponds to public betting addresses.**



Figure: One of games

Case Studies-TRON



Figure: TRX_USDT

Known Addresses	Details
TGDa2DmBR1i2ofDBowfcjcgweyvpg9DDyU	幸运哈希
TTwC4KJAGPwHvsde5p6MxgksZMcbeM1Swm	大小单双
THRy4fj3Xbhdah21VFZQU8a2sFoVwpmDrL	
TBaoBhNp2wBHrqPNcwK9cLf1gNmPXdY6dD	牛牛游戏
TBZEkbueuW3Vhy3hJJ2UTCGtCMZydiHrez	组合游戏
TFJvYPSxbq4GBcTak14N8ZEHHnpTUwZ3u4	
TYfJNiaRPMLsRQnKZrC11y5D4YSez7RknX	
TAtMZegrXPY0amF8hSMhv9J3G5Z9BZz1R3	
TWt2bNRNas6pbekd4LHb2EcooWZCCDdofj	龙虎和
THqAUKa977Wc1doq9mPjne126fjbfQTCee	
TVpxm5XhL4C415jASUF8uco2tv52f626n2	
TFucXxbfHcQg4abbUdXU1PBodB4xWvfVSz	百家乐
TBqFXt28uU7uXuvSdzWffGa8CtnUrZyj4G	
TYLGg1Kec54U52VPy68TSAPyqhaCyLVVfi	
TQpLsV7reeGc6aeU9tgVqDA99Q52eMMMMM	兑换地址

Figure: Known betting and exchange addresses

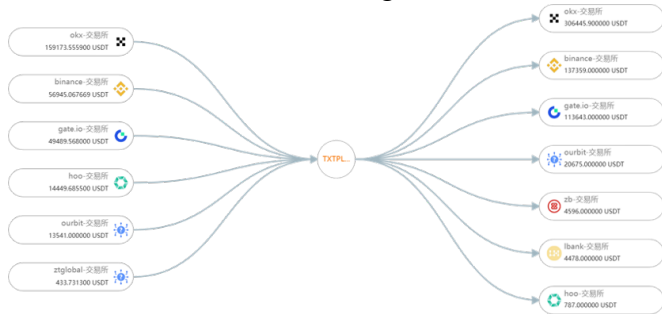
Case Studies-TRON

Table: Normalization values of $\tilde{N}_v^{\text{in}}$ and $\tilde{N}_v^{\text{out}}$ for partial key addresses, along with corresponding roles. *Bonus payers*, serving as intermediaries, have more output transactions than input ones. *Exchanges* receive a relatively high volume of input and output transactions.

Address	$\tilde{N}_v^{\text{in}}$	$\tilde{N}_v^{\text{out}}$	Role
TVXn6N...tXXXXXX	0	0.863	Bonus payers
TDDRAf...D888888	0	0.324	Bonus payers
TQpLsV...eMMMMMM	0	0.089	Bonus payers
TPRKHn...Zibb75	0.053	0.049	OKEEx Exchange
TQ6VZ7...KiSMeV	0.023	0.025	OKEEx Exchange
TV866T...PoEfFJ	0.016	0.021	Binance Exchange
TEvgA6...9Z31J7	0.013	0.035	Binance Exchange
TFfEJf...CR7qgh	0.017	0.020	Gate.io Exchange
TJHSgQ...z1rU7u	0.005	0.006	Gate.io Exchange

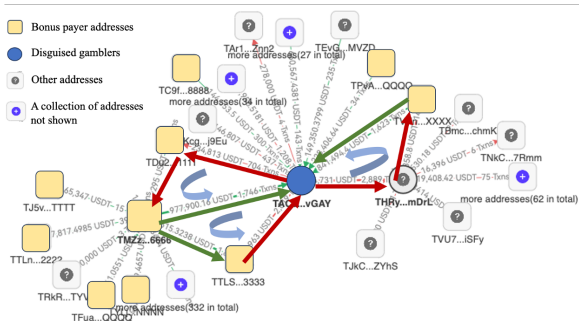
Case Studies-TRON

- Funding sponsors:** Since every transaction on the TRON system requires a certain amount of TRX as initial gas fees originated from funding sponsors for bonus payers. In this case, one of the suspected funding sponsor addresses(i.e. TXTPLF...Y82zGH) **transferred 500 TRX to the bonus payer address** (i.e. TVXn6N...tXXXXX) on July 27, 2022, at 23:00. Evidently, the sponsor address and bonus payer address belong to the same project team. The sponsor address has multiple deposit and withdrawal interactions with exchanges like Binance, OKEx, and Gate.io.



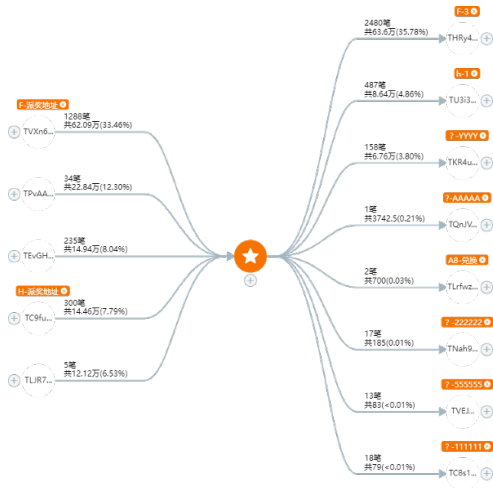
Case Studies-TRON

- **Disguised gamblers:** Among gamblers, some suspected addresses transfer funds to bonus payers. Such gamblers are highly likely to be delegates controlled by project members. In this case, suspected disguised gambler addresses(i.e., TYe2Kt...DRouGL, TACJi9...anvGAY) exhibit the following characteristics: **(1) Disguised gamblers send deposits to gambling betting addresses. (2) Bonus payers send rewards to disguised gamblers. (3) Disguised gamblers return profits to bonus payers.**



Case Studies-TRON

- Real gamblers:** Gambling rules reveal that addresses that transfer funds to the betting address and receive partial funds back from bonus payer addresses are likely to be real gambler addresses(i.e., TUuvLo...eCGf4Z, TR5VMU...C86PUm). **It only transfers funds to known betting addresses and known bonus payer addresses transfer funds to this address.**



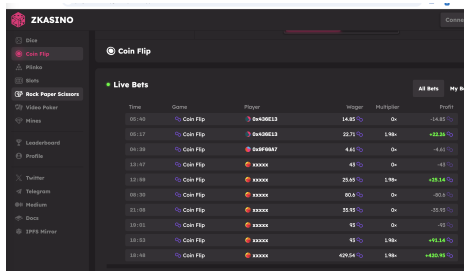
Case Studies-TRON

Table: Account statistical features of different roles on TRON Gambling, including the number of input transactions(*NIT*), the number of output transactions(*NOT*), the total amount of input transactions(*TAIT*)[Unit: USDT], the total amount of output transactions(*TAOT*)[Unit: USDT].

Addresses	Roles	<i>NIT</i>	<i>NOT</i>	<i>TAIT</i>	<i>TAOT</i>
TVXn6N...tXXXXXX	Bonus payers	366	11,539	5,142,813.9	5,142,813.3
TXTPLF...Y82zGH	Funding sponsors	3,654	3,755	10,020,873.5	10,040,463.5
TYe2Kt...DRouGL	Disguised gamblers	6,947	9,494	14,299,163.9	14,298,197.1
TUuvLo...eCGf4Z	Real gamblers	11	19	530.54	530.51
TGDa2D...g9DDyU	Betting addresses	81	3	1,216.8	1,199.9
TQpLsV...MMMMM	Exchange addresses	87	4	1,994.9	1,991.9

Case Studies-Arbitrum

- *ZKasino* is a popular casino platform mainly operating on Arbitrum and Polygon.
- Instead, players retain full control of their funds in their own wallets, placing bets directly without any intermediary involvement.
- With more rapid transaction speeds and lower fees, ZKasino is more popular.
- Each game on the platform corresponds to a smart contract address, including *Dice*, *Coin Flip*, *Plinko*, *Slots*, etc.
- We select *Coin Flip Game* address(0xC4A4...E5279c) as the source node and extract its multi-layer transfers.



The screenshot displays the ZKASINO interface for the Coin Flip game. On the left is a sidebar with navigation links: Dice, Coin Flip (selected), Plinko, Slots, Rock Paper Scissors, Video Poker, Mines, Leaderboard, Profile, Twitter, Telegram, Medium, Docs, and 1PPS Miner. The main area is titled 'Coin Flip' and shows 'Live Bets'. There are two tabs: 'All Bets' and 'My Bets'. A table lists the following bets:

Time	Game	Player	Wager	Multipier	Profit
05:40	Coin Flip	0x436E13	14.85	0x	-14.85
05:17	Coin Flip	0x436E13	22.71	1.98x	+22.36
04:39	Coin Flip	0x6F06A7	4.61	0x	-4.61
13:07	Coin Flip	0XXXXX	43	0x	-43
12:09	Coin Flip	0XXXXX	25.95	1.98x	+25.14
08:30	Coin Flip	0XXXXX	80.6	0x	-80.6
21:08	Coin Flip	0XXXXX	35.93	0x	-35.93
19:01	Coin Flip	0XXXXX	93	0x	-93
18:53	Coin Flip	0XXXXX	93	1.98x	+91.14
18:48	Coin Flip	0XXXXX	429.54	1.98x	+420.49

Case Studies-Arbitrum

- We first identified *Uniswap, SushiSwap, Paraswap, Arbswap Exchanges* with both high $\tilde{N}_v^{\text{in}}$ and $\tilde{N}_v^{\text{out}}$ (i.e., 0x1B02...997506, 0x68B3...65FC45)
- Most likely gamblers with high $\tilde{N}_v^{\text{out}}$ (i.e., 0x2818...C0D2C7, 0xB28F...C9B282E) directly linked to the Coin Flip Game contract address.
- We discovered new delegates in Arbitrum casinos beyond those found on TRON.
 - **Cross-chain Bridges:** Cross-chain Bridges (i.e., 0x80C6...99BCF8, 0xE4ED...00BCE8), involved in large volume transfers across networks, help gamblers transfer ETH back and forth between the Ethereum mainnet and Arbitrum.
 - **Liquidity Providers:** *Regular Gamblers* or *Casino Stakeholders* tend to use diverse *Liquidity Providers* such as liquidity pools (i.e., 0x1054...ADE261, 0x652D...AC6D97), staking platforms (i.e., 0xEA8D...4C2176, 0x1F80...E58A28), and lending protocols (i.e., 0x2032...616B1F, 0xF4B1...0659E1) to exchange ETH for tokens that can be used for gambling. These *Liquidity Providers* handle frequent but smaller transactions. This is consistent with the staking-related activity and updates officially released by *Zkcasino*.

Model Evaluation, Ablation Study and Parameter Analysis

Ablation Studies: Without graph learning (0 on the horizontal axis), as well as the performance of LSV (1), TSV (2), GSV (3), and FSV (4) views within graph learning

Table 4: Model evaluation(%) of our approach and baselines on Ethereum, TRON, and Arbitrum

Baselines	Ethereum			TRON			Arbitrum		
	Purity	ACC	NMI	Purity	ACC	NMI	Purity	ACC	NMI
SuperGAT+KMeans	75.00	68.75	67.06	70.59	58.82	64.75	69.23	53.85	53.20
EGC+KMeans	68.75	25.00	28.96	64.71	47.06	56.22	61.54	46.15	54.60
GATv2+KMeans	75.00	68.75	78.35	76.47	64.71	71.21	76.92	46.15	68.78
GNN-FiLM+KMeans	68.75	50.00	65.18	76.47	64.71	66.38	53.85	53.85	48.92
DirGNN+KMeans	62.50	43.75	64.85	70.59	58.82	62.48	76.92	61.54	58.94
SuperGAT+DBSCAN	68.75	62.50	65.52	70.59	64.71	67.55	61.54	61.54	54.11
EGC+DBSCAN	50.00	31.25	51.26	76.47	58.82	64.90	69.23	46.15	40.05
GATv2+DBSCAN	81.25	75.00	68.47	76.47	76.47	73.89	61.54	61.54	65.30
GNN-FiLM+DBSCAN	56.25	43.75	60.28	64.71	47.06	56.75	69.23	61.54	59.56
DirGNN+DBSCAN	75.00	56.25	67.06	70.59	58.82	67.90	76.92	69.23	73.03
RoIX	68.75	62.50	42.98	64.71	52.94	55.92	61.54	46.15	42.20
EPLCD	68.75	50.00	59.26	76.47	58.82	64.30	69.23	69.23	64.65
Ours	81.25	81.25	79.66	82.35	82.35	78.34	76.92	76.92	71.52

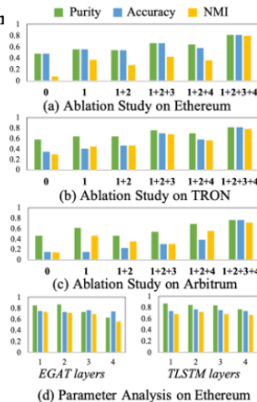


Figure 5: Ablation studies and parameter analysis

Implication

- **Theoretical Implication**

- Unlike address-level detection methods, our role-level identification approach assigns specific roles to each node, without relying on predefined labels.
- Our Method can extend to blockchain platforms utilizing the Ethereum Virtual Machine account model.
- It shows near-linear time complexity when processing large, sparse blockchain ecosystems.

- **Practical Implication**

- This study elaborates on distinct role compositions and operation mechanisms across the most mainstream blockchain ecosystems. **Ethereum casinos** closely resemble centralized casinos, and **TRON casinos** and **Arbitrum casinos** conduct on-chain transactions for exchange, betting, and bonus payment.
- For regulatory bodies, accurately identifying key and hidden figures helps disrupt illegal gambling activities more effectively.
- For regular participants, understanding platform roles reduces information asymmetry, protecting them from manipulation.

Future Work

- To include **data** from emerging blockchain systems like Polygon and EOS for broader insights.
- To increase our focus on **non-EVM platforms**, such as converting the UTXO model to an account-based model using heuristic address clustering.
- To develop a hybrid role identification method based on structural entropy in overlapping community scenarios. (Entities behind addresses may assume multiple roles, and these roles may **evolve over time**.)
- To explore richer **transactional semantic information of smart contracts** to provide a more comprehensive network analysis.

Conclusion

- We *first* presents a systematic investigation into the **security and transparency** of crypto casinos through the lens of role-level identification.
- We propose an **unsupervised dual-stage role identification** algorithm that combines key node identification based on **1D structural entropy** and a self-supervised GNN-based framework enhanced with **2D global structural entropy minimization**.
- Extensive experiments are conducted on real-world cases on Ethereum, TRON, and Arbitrum to uncover diverse role and their collusion schemas among roles.



Paper

Gamblers or Delegates: Identifying Hidden Participant Roles in Crypto Casinos

Jiaxin Wang¹, Qian'ang Mao¹, Hongliang Sun², Jiaqi Yan^{1*}

1. School of Information Management, Nanjing University, China

2. University of Nottingham Ningbo China

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